

# Keystroke/Mouse Usage Based Emotion Detection and User Identification

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**Abstract**—Emotions are primarily thought of as mental experiences of body states, which are mostly shown in the face with precise and specific muscle patterns. It is, perhaps, the most critical attribute of living beings, and is extremely difficult to detect and generate artificially. Its detection always remains a well-explored classical problem. Existing approaches for detecting human emotions generally demand significant infrastructural overheads. Excluding these overheads, in this paper, we propose a much simpler way of emotion detection. To do so, We have induced different states of emotion through different multimedia components, and then collected participants' keystrokes (free text) and mouse usage data through a custom-developed survey. We have used several existing classifiers (KNN, KStar, RandomCommittee and RandomForest) and a newly proposed light-weight classifier namely Bounded K-means Clustering, to analyze those usage data for different emotional states. Our analysis demonstrates that emotion can be detected from the usage data up to a certain level. Moreover, our proposed classifier enables the best detection of five emotional states namely happiness, inspiration, sympathy, disgust, and fear compared to other existing classifiers. Besides, the analysis also reveals that user identification through usage dynamics does not result in a good level of accuracy when usage gets influenced by different emotional states.

**Index Terms**—Keystroke Dynamics, Bounded K-means Clustering, Dominant Emotions, KNN.

## I. INTRODUCTION

Emotionally-aware system is the first step to the world of human computer (cyborg) era. Systems capable of detecting emotions can fulfill the dream of automated interaction in social media. There are numerous such applications of emotion detection. It can bring evolutionary changes in diversified fields such as medication, E-learning, entertainment, marketing, etc. It can play a big role in case of psychotherapy using intelligent tools. Besides, research has shown that 30% - 50% of time spent on a computer is wasted due to having frustrating experience [1]. Emotionally aware system can be handy to mitigate such cases.

To recognize, interpret, process, and simulate the human affects, study and development of systems are required. Such study and development are called affective computing [2].

The main purpose of this computing is to expound emotional states of humans and adapt its behavior giving an appropriate response for those emotions. Here, to recognize emotional information it is required to extract meaningful patterns from different types of data gathered from users. Existing procedures for capturing data about users' physical states or behavior mostly use passive sensors such as video camera [17] for detecting facial expressions along with body posture and gestures [9], microphone for capturing speeches [14], body sensors for gathering physiological data such as skin temperature, etc. All these approaches exhibit significant accuracy, however, demand additional expensive equipments and hardware setup.

On the other hand, in our study, we have proposed an inexpensive emotion detection model excluding all the additional hardware overheads. Here, we have used simple keystroke and mouse operation dynamics to gather usage data from users. The keystroke dynamics [3] or typing dynamics refer to the automated method of identifying or confirming the identity of an individual based on the manner and the rhythm of typing on a keyboard. Thus keystroke dynamics provide a behavioral biometric. The rhythm, with which characters are typed can be different for different emotional states. This logic galvanized our emotion detection study.

In our study, first, we have used multimedia components to induce different emotional states. As films were successfully used by other researchers for inducing emotional states in the past [4], we have designed a survey system including different multimedia clips to perform the emotion induction phase. After each clip, we have presented a set of questionnaire for collecting keystroke information. We have also taken different mouse events into consideration. Then we have exploited different classifiers such as KNN and used the WEKA [5] tool on the collected usage data for classifying different emotional states. Additionally, we propose a new classifier namely Bounded K-means Clustering and employ it for the similar task. We point out efficacy of our proposed classifiers through presenting its accuracy in the detection

tasks. In addition to emotion detection, we have also analyzed the usage data for user identification in different emotional states. Note that earlier studies [24] in the literature have pointed out that users can be identified through analyzing the usage data. Going beyond this finding, our study points out how far we can go if multiple emotional states get blended while collecting the usage data intended for user identification.

The following list presents a set of our contributions made in this paper.

- We have developed a survey system for collecting usage data in different emotional states. We have utilized the system to collect usage data from 35 participants.
- We have attempted to classify the emotional states in various forms from the usage data exploiting several existing classifiers.
- We propose a new light-weight classifier namely Bounded K-means Clustering and employ it in the classification task. The attribute of being light-weight has been confirmed through real experimentation.
- Finally, we have attempted to identify different users from the usage data collected under different emotional states. Here, we demonstrated how blending of different emotional states while collecting usage data exhibits significant impacts on user identification.

The contents of this paper are organized as follows: Section 2 briefly presents related work and motivation behind our study, Section 3 contains methodology of our study where we elaborate experimentation and outcomes, Section 4 presents a summary of findings of our study. Finally in Section 5, we discuss our future work and conclusion.

## II. RELATED WORK AND MOTIVATION

Emotion is a complex attribute of human being, which depends on various factors such as time, space, context, and individuals. Therefore, finding a universal method for detecting emotion is a difficult task to do. Even then several studies in the literature have attempted for building emotion detection models. Most of the studies [19], [18] used self-reported emotional states and went through emotion detection procedures only based on the self-reported states. On the contrary, in our study, we have attempted to induce different emotional states first, and then analyzed usage data to detect the emotional states. Therefore, in the following subsections, we briefly discuss related research studies dividing the topic into two subgroups-emotion induction and emotion detection. Afterward, we present a brief overview on user identification, and then we present motivation of our study.

### A. Emotion Induction

Different emotional states can be induced through several methods such as imagination, music, hypnosis, social interaction, imitation of a facial expression, or memories of positive and negative life events [8], etc. Research studies used different combination of these methods to get effective induction. We briefly present a summary on it below.

1) *Velten Procedure*: Velten procedure [10] is one of the most used approaches to induce a given mood. In this procedure, participants are given a number of statements to induce a specific mood. However, it has the limitation of achieving induction not lasting more than a very short period of time.

2) *Emotion Induction Exploiting Films*: For longer lasting emotional state or mood induction, the use of film clips including sounds has been proven to be handy. It has been tested in both laboratory and field research and found to produce salient and enduring moods [9]. Besides, this usage works best when subjects (not studying psychology) are treated individually and are introduced to the purpose of the study. Many researchers used film clips to successfully induce different emotional states. One of the pioneer studies in this regard is from Westermann's et al. [11], which attempted to find films that induce different emotional states (dimensional space). Besides, J. Gross et al. [12] attempted to find as many films as possible to elicit discrete emotions and find the best films for each discrete emotional state. Zimmerman et al. 2003 [13] used films of around 9-10 minutes length for each mood states to induce emotions. In their work, they have shown promising results in inducing emotion via films.

### B. Emotion Detection

1) *Using Hardware*: Existing emotion detection approaches usually demand additional infrastructures such as web-cam, body mounted hardware, etc., which are often intrusive and expensive. This approaches also require specialized information such as voice [14], thermal images of faces [15], gestures [9], facial expression [17], etc., which are not always available.

2) *Using Keystroke Dynamics*: The solution to the problem discussed above is using the analysis of rhythm of typing pattern in a standard keyboard i.e. keystroke dynamics. Keystroke dynamics includes the study of keyboard typing pattern of individuals while using computer. It also includes extracting keystroke timing features such as the typing speed, duration of key press, time between two key press etc. Several studies are done for detecting emotion via keystroke dynamics. Clayton Epp et al. [18] gave a solution to determine user emotion by analyzing keystroke dynamics. They conducted a field study and collected user's keystrokes and self reported emotional states. They attempted to classify 15 different emotional states. Their top results include 2-level classifiers for confidence, hesitation, nervousness, relaxation, sadness and tiredness with accuracies ranging from 77% to 88%. Preeti Khanna [15] presented a method to recognize emotion via keystroke dynamics where they considered neutral, positive and negative emotional states. They showed that users are prone to make mistakes followed by backspace and unrelated keys in negative emotional states. Besides, they also found that typing speed of users reduces when they are in negative emotional state. They have used various classification algorithms - simple logistics, SMO, Multilayer Perception, Random Tree, J48, BF Tree and WEKA [5] tool for classification. Vizer et al. [10] used keystroke timing features of free text along with linguistic features to classify cognitive and physical stress. Their accuracy was 63%

for cognitive stress, though they stated that their solutions should be tested with varying typing abilities, keyboards, physical and cognitive abilities and stressful situations.

3) *Using Text Pattern analysis*: A recent study [16] showed that a chat system where textual message is analyzed to extract different emotional states of the user. Advanced keyword-spotting technique was used to recognise the affective content of the textual message. For detecting affective meaning syntactical sentence-level processing was applied. This analogy inspired another recent work [19] where it has been shown that combined method of keystroke dynamics and text pattern analysis improved accuracy of emotion detection, rather than using them separately. Their work performed well with fixed text than free text. They have used self reported emotion to train their system.

### C. User Identification

Most of the user identification or authentication systems are based on passwords or biometric data. One of the most concerning facts about these password based authentication systems is that passwords can be guessed or stolen [20]. The overall methodologies for electronics device user identification according to the study [21] can be classified as a) knowledge based (password, PIN), b) object based, which based on possession of a token, c) biometric based, which relies on the physical characteristics of user. However, different studies [22], [23], [20] show that all of the procedures stated above is vulnerable and prone to security threat such as eavesdropping, physical spoof exploiting standard biometric templates. Therefore, to address these issues, user authentication using keystroke dynamics has come into play. Keystroke analysis have been successfully used in user authentication [24], where users are recognized successfully by their typing pattern along with their correct user name and password. Classification algorithms for the analysis of keystroke dynamics for user authentication include neural networks [26], distance measures [24], [27], decision trees [28], and other statistical methods [27], [29], [30]. Due to the differences in data collection approaches and classification methods, a comparison of performance across studies is difficult [29]. Accuracy of user authentication using keystroke rests on two vital things 1) standard of keyboard used and 2) emotional state of the user. In our study, we have analyzed this vulnerability of user authentication system using keystroke dynamics for different emotional states.

### D. Motivation Behind Our Study

Most of the authentication systems and emotion detection systems [19], [18], [15] are based on keystroke analysis of fixed text, where users are prompted to type a specific paragraph or text. They did not induce emotion in user before collecting keystrokes of user. They have taken self reported emotional states into consideration.

In most cases, using fixed texts showed better results than using free text. However, the problem with fixed text is that it does not relate to real time scenario. We hope our system

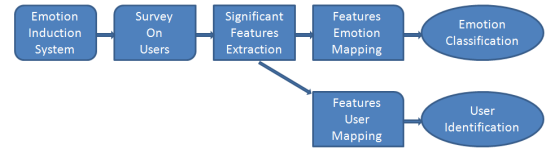


Figure 1: Our proposed approach for emotion detection and user identification

to be able to detect our emotional state from our interaction with it and generally we do not interact using fixed text pattern rather we use free text. Therefore, a model that uses free text for keystroke dynamics in detecting emotion, will be far more useful and desirable.

In case of self reporting, it lacks variation of emotions. The bigger problem is, while the reporter is in negative emotional states or emotionally down, it is very unlikely that he or she will participate in a survey.

Furthermore, keystroke dynamics alone is not fruitful in detecting some emotions such as surprise, amusement [31]. Therefore, we have also included mouse events as attributes for the analysis.

## III. METHODOLOGY OF OUR STUDY

We have conducted our study in five sequential phases. First, we have developed an emotion induction system. Our developed system is basically a survey system where we use findings already available in the literature to induce emotions, which is multimedia components. The survey system also captures after effects of emotion induction through letting a user to type free text at own will. Second, we present our developed system to several users who interact with the system. Third, we analyze usage data obtained through the interaction between users and our developed system. Our analysis extracts significant features from the user data. Fourth, we map the extracted user data to different emotional states. Finally, we exploit the extracted features to classify different emotions. We also have analyzed the extracted data to identify users, first mapping those data to different users. We elaborate each of the phases in detail in the following subsections.

### A. Emotion Induction System

Emotional States can be described by three basic emotional dimensions:

1. A dimension of affective valence (also called pleasure) ranging from positive (pleasant) to negative (unpleasant).
2. A dimension of arousal, ranging from calm (low-arousal) to excited (high-arousal).
3. A dimension of dominance or control, ranging from controlled to in control.

We have used 9 video clips (around 2 minutes long) of these different dimensions of emotions to induce emotion into users as depicted in Table I. There were related still pictures and texts also.

It can be possible that one particular video can induce multiple emotional states. However, we have selected the

| Emotion     | Corresponding video clip                          | Length (min) | Category |
|-------------|---------------------------------------------------|--------------|----------|
| Amusement   | Mr. Bean taking the stairs [32]                   | 1:57         | PVLA     |
| Happiness   | Baby laughing hysterically at ripping paper [33]  | 1:20         | PVHA     |
| Inspiration | Heartwarming Thai commercial [34]                 | 1:52         | PVLA     |
| Surprise    | Street magic of Dynamo (a magician) [35]          | 2:24         | PVHA     |
| Sadness     | Story of a blind father and his daughter [36]     | 2:01         | NVLA     |
| Sympathy    | Sufferings of famine in Somalia [37]              | 2:01         | NVLA     |
| Anger       | Rape during the liberation war in Bangladesh [38] | 1:31         | NVHA     |
| Disgust     | Man vs. Wild - eating giant larva [39]            | 1:44         | NVHA     |
| Fear        | He dies at the end (scary video) [40]             | 2:30         | NVHA     |

Table I: Video clips used for eliciting emotions. Here PVLA-Positive Valence Low Arousal, PVHA-Positive Valence High Arousal, NVHA-Negative Valence High Arousal, NVLA-Negative Valence Low Arousal

videos in such careful manner that one particular video induces one specific emotion in most of the cases. All the videos, still pictures and texts were presented among 10 persons and they were asked to tell their emotional states after watching each video, still picture and text. Then we selected the best set of videos for 9 different emotional states (amusement, happiness, inspiration, surprise, sadness, sympathy, anger, disgust and fear). There was no video for neutral state of emotion. We considered the users to be in a neutral state when they started the survey and filled up the basic information.

In our induction and survey system, we presented these selected videos, images and texts to induce certain emotions and then prompted the user with some related questions. The answers to the questions were 50 to 100 characters long. We captured keyboard typing parameters from this answer. There were also a paragraph for each emotional state, for which survey participants' used mouse scrolling. There were some multiple choice questions, for which participants' used the mouse clicks also. Therefore, from the survey we captured individual user's keyboard typing parameters, mouse click and scrolling parameters.

### B. Survey on Users

We conducted survey on computer users of various age range, occupations and computer skill. There were a total of 35 participants. 28 of them are male and 7 are female. Most of them are students, some are software engineers and the rest are from different occupations. 92% of the users of the survey has good typing skill. Fig. 3 shows the demography of the participants.

### C. Significant Features Extraction

In the user survey, while answering the questions, user's certain usage attributes were logged automatically in every 5 seconds. There were 17 attributes that were logged. The attributes are average mouse left click, average mouse right click, average mouse double click, average mouse scroll, average cursor x-distance, average cursor y-distance, average key down to up time, average key up to down time, average key down to down time, average regular key press, average enter key press, average arrow key press, average backspace key press, average function key press etc. We also collected some meta data from the user. These were the name, age, occupation and birth year of the survey participant. These meta data were later used for user identification as neutral emotional state data.

### D. Emotion Classification

We exploited the survey data using different classifiers. Our main focus was to classify users' emotional states. Besides, we also attempted to classify user's identity.

1) *Classifiers Used:* There exists several classifiers serving different purposes. The following classifiers and tools were used for data analysis.

a) Bounded K-means Clustering:

Conventional K-means Clustering [6] aims to classify every data point into an existing cluster based on nearest centroid. However, in case of our study, we observed that usage data is confined in a limited region. Therefore, we introduced a modified version of Conventional K-means Clustering namely Bounded K-means Clustering to work only within a limited range of data. In this newly proposed approach, we put a constraint on the cluster size (C). Here, centroid of the cluster is the mean value of the data and the cluster size (C) is set to a weighted sum of average and standard deviation as follows:

$$C = (AVG \times W_{avg}) + (SD \times W_{sd}) \quad (1)$$

Here,  $W_{avg}$ ,  $W_{sd}$  represent weight of average value (varied from 0% to 100%) and weight of standard deviation (varied from 0% to 500%) respectively. The cluster size (C) was varied to find out the optimal cluster size so that the total false acceptance and false rejection are minimized. The number of clusters was fixed beforehand based on the number of emotional states we classified. Fig. 4 depicts a scenario of the proposed Bounded K-means Clustering method. Here, we have considered two attributes (Average mouse click and Average arrow click) from the usage data of four emotional states (Anger, Happiness, Disgust and Fear). Each circle represents a cluster of a specific emotion.

b) KNN (K Nearest Neighbors) [7]:

We used this well known classifier where a test point is classified by examining its neighbor points. We varied both the number of neighbors and the number of attributes.

c) Weka [5]:

Weka is a collection of machine learning algorithms for data mining tasks. We used the Weka tool for our classification. We used Bagging, IB1, KStar, RandomCommittee, RandomForest,

Name (optional):

Birth Year:

Occupation:

Gender:  
☒ Male  
☐ Female

Computer Operating Skill:  
☒ Beginner  
☐ Intermediate  
☐ Advanced

(a) Start page of the survey system

Read and scroll down to the end of this paragraph. Click next button at the end

On the very first day, God created the cow. God said "You must go to the field with the farmer all day long with the farmer and I will give you a life span of sixty years." The cow said, "That is kind of a tough life... you want me to live for sixty years! Let me have twenty years instead". And, God agreed. On the second day, God created the dog. God said, "Sit all day by the door of your house and bark at anyone who comes. I will give you a life span of twenty years." The dog said, "That is too long to be barking

(b) Text presented to participants for mouse scrolling

\*\* Please write something (within 50 words) about your favourite comedian.

Do you like comedy movies?  
☐ Yes  
☐ No  
☐ Won't disclose

Do you have any pet?  
☐ Yes  
☐ No  
☐ Won't disclose

Who is your favourite comedian?

(c) Questionnaire presented after inducing each type of emotions

Figure 2: A graphical presentation of the survey application

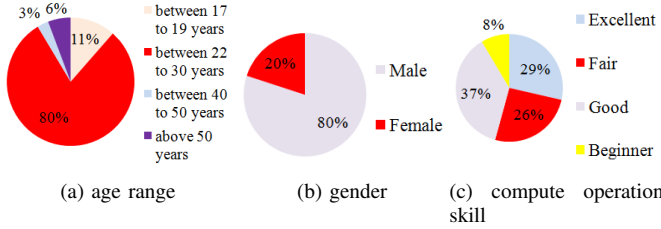


Figure 3: Category of users based on age range, gender and computer operating skill

| Method          | FP (%) | FN (%) | Accuracy (%) |
|-----------------|--------|--------|--------------|
| Bounded k-means | 43     | 34     | 23           |
| Knn             | 76     | 0      | 24           |
| Bagging         | 68     | 0      | 32           |
| KStar           | 67     | 0      | 33           |
| RandomCommitte  | 67     | 0      | 33           |
| Random Forest   | 65     | 0      | 35           |
| Random Tree     | 71     | 0      | 29           |
| Majority Voting | 85     | 0      | 15           |

Table II: Bounded k-means Clustering, Knn and Weka tool classifiers for Ten class emotion classification (here, FP and FN refer to False Positive and False Negative respectively)

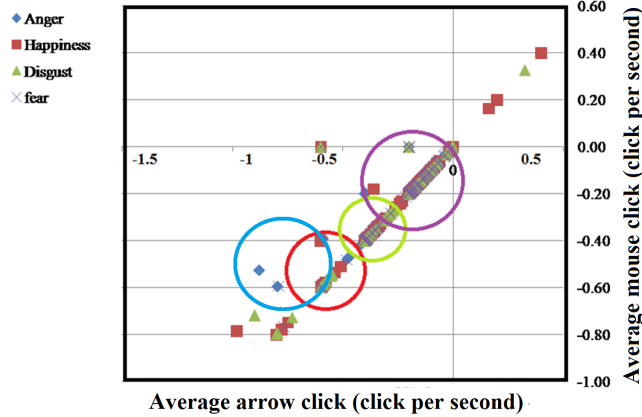


Figure 4: Proposed Bounded K-means Clustering method for emotion classification

RandomTree classifier over the data using Weka.

2) *Approaches For Classification:* At first we tried to classify human emotions based on the usage attributes captured in each step of the survey process. Every row of the captured data is comma separated where the last two tags are 'emotion label', 'user label' respectively. For emotion classification, we

at first, normalized all the survey data. We used the term 'False Acceptance Rate'/'False Positive'to define the percentage of emotional data identified as wrong emotional state. And 'False Rejection Rate'/'False Negative'implies percentage of emotional data that can't be classified into any emotional state.

#### a) Ten Classes of Emotions

We tried to classify 10 different emotional states. These are amusement, happiness, inspiration, surprise, sadness, sympathy, anger, disgust, fear and neutral emotion. We tried different classification approaches: Bounded k-means clustering, KNN and WEKA [5] tool classifiers. Our classification result is depicted in Table II and Fig. 5. Here, we observed that a single sample data is not classified in same class by different classifiers. Therefore, we attempted to classify user's emotion using a majority voting classifier to find if the accuracy improves or not. However, running the classifier we found that the accuracy did not improve.

#### b) Two Classes of Emotions

We grouped the 10 different emotion classes into 2 groups. The positive valence (pleasant) emotions such as amusement, happiness, inspiration and surprise were grouped into one

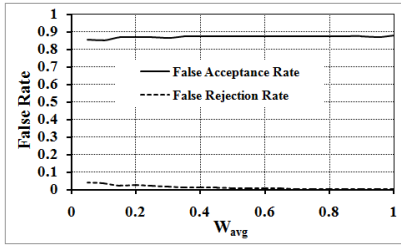


Figure 5: Percentage of average versus false acceptance rate and false rejection rate in Bounded k-means clustering for 10-class emotion classification

| Method          | FP (%) | FN (%) | Accuracy (%) |
|-----------------|--------|--------|--------------|
| Bagging         | 38     | 0      | 62           |
| IB1             | 42     | 0      | 58           |
| KStar           | 38     | 0      | 62           |
| RandomCommittee | 39     | 0      | 61           |
| Random Forest   | 38     | 0      | 62           |
| Random Tree     | 39     | 0      | 61           |
| Bounded K-means | 41     | 10     | 49           |

Table III: Weka tool classifiers and Bounded k-means Clustering for 2-class emotion classification (using original data)

class. The negative valence (unpleasant) emotions such as sadness, sympathy, anger, disgust and fear were grouped into another class. The result of running different classifiers from WEKA [5] tool and Bounded K-means Clustering on 2-classes of emotions data is depicted in Table III, IV, V, and VI.

In Table III, we can find the result found after running different classifiers on the original usage data. For further improvement, we form a set of new data points (1<sup>st</sup> derivative of usage data) by considering the distance between each original data point and their mean. Similarly, we populated 2<sup>nd</sup> derivative of data from 1<sup>st</sup> derivative of data and 3<sup>rd</sup> derivative of data from 2<sup>nd</sup> derivative of data. We can find from the table that accuracy has increased as we took further derivative. The best accuracy 69% was achieved for 3<sup>rd</sup> derivative data after running classifier RandomCommittee of WEKA [5].

#### c) Dominant Classes of Emotions

From the unsupervised Bounded k-means clustering of 10 emotions, we found 10 different clusters, however this clusters were not dominated by only one class of emotion. Rather, we found that there are some emotions, which dominates not only in his own cluster but also in the clusters of others. Table VII shows a matrix representation where each row resembles dominance of different clusters on a specific cluster. Therefore, We attempted classifying dominant emotions only (amusement, surprise, sadness, and anger) using Bounded K-means Clustering and WEKA tool classifiers. The result is depicted in Table VIII and Fig. 6.

#### d) Less Dominant Classes of Emotions

From the cluster matrix we can identify the clusters (emotions), which has less influence over their own clusters and

| Method          | FP (%) | FN (%) | Accuracy (%) |
|-----------------|--------|--------|--------------|
| Bagging         | 38     | 0      | 62           |
| IB1             | 46     | 0      | 54           |
| KStar           | 39     | 0      | 61           |
| RandomCommittee | 37     | 0      | 63           |
| Random Forest   | 39     | 0      | 61           |
| Random Tree     | 41     | 0      | 59           |
| Bounded K-means | 39     | 12     | 49           |

Table IV: Weka tool classifiers and Bounded k-means Clustering for 2-class emotion classification (Using 1<sup>st</sup> derivative of data)

| Method          | FP (%) | FN (%) | Accuracy (%) |
|-----------------|--------|--------|--------------|
| Bagging         | 37     | 0      | 63           |
| IB1             | 43     | 0      | 57           |
| KStar           | 36     | 0      | 64           |
| RandomCommittee | 34     | 0      | 66           |
| Random Forest   | 36     | 0      | 64           |
| Random Tree     | 38     | 0      | 62           |
| Bounded K-means | 40     | 10     | 50           |

Table V: Weka tool classifiers and Bounded k-means Clustering for 2-class emotion classification (Using 2<sup>nd</sup> derivative of data)

| Method          | FP (%) | FN (%) | Accuracy (%) |
|-----------------|--------|--------|--------------|
| Bagging         | 34     | 0      | 66           |
| IB1             | 38     | 0      | 62           |
| KStar           | 36     | 0      | 64           |
| RandomCommittee | 31     | 0      | 69           |
| Random Forest   | 31     | 0      | 69           |
| Random Tree     | 33     | 0      | 67           |
| Bounded K-means | 34     | 32     | 34           |

Table VI: Weka tool classifiers and Bounded k-means Clustering for 2-class emotion classification (Using 3<sup>rd</sup> derivative of data)

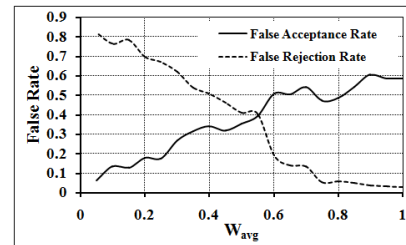


Figure 6: Percentage of average versus both false acceptance rate and false rejection rate in Bounded k-means for dominant emotion classification

| Emotion Classes | Amusement | Happiness | Inspiration | Surprise | Sadness | Sympathy | Anger | Disgust | Fear |
|-----------------|-----------|-----------|-------------|----------|---------|----------|-------|---------|------|
| Amusement       | 23        | 5         | 15          | 13       | 11      | 14       | 9     | 2       | 5    |
| Happiness       | 19        | 12        | 13          | 15       | 10      | 7        | 15    | 7       | 10   |
| Inspiration     | 19        | 9         | 16          | 13       | 9       | 6        | 5     | 5       | 7    |
| Surprise        | 14        | 15        | 8           | 29       | 11      | 8        | 19    | 7       | 2    |
| Sadness         | 19        | 15        | 7           | 11       | 22      | 0        | 4     | 4       | 4    |
| Sympathy        | 17        | 10        | 7           | 12       | 13      | 9        | 16    | 9       | 6    |
| Anger           | 13        | 6         | 15          | 16       | 12      | 10       | 19    | 5       | 4    |
| Disgust         | 13        | 10        | 11          | 12       | 10      | 11       | 12    | 11      | 10   |
| Fear            | 16        | 11        | 10          | 13       | 9       | 9        | 12    | 8       | 12   |

Table VII: Matrix of domination for 9-emotion classes (without the neutral emotion (numeric values))

| Method          | FP (%) | FN (%) | Accuracy (%) |
|-----------------|--------|--------|--------------|
| Bagging         | 49     | 0      | 51           |
| IB1             | 58     | 0      | 42           |
| KStar           | 54     | 0      | 46           |
| RandomCommittee | 48     | 0      | 52           |
| Random Forest   | 47     | 0      | 53           |
| Random Tree     | 56     | 0      | 44           |
| Bounded K-means | 48     | 5      | 47           |

Table VIII: Weka tool classifiers and Bounded k-means Clustering for dominant emotion classification

| Method          | FP (%) | FN (%) | Accuracy (%) |
|-----------------|--------|--------|--------------|
| Bagging         | 62     | 0      | 38           |
| IB1             | 69     | 0      | 31           |
| KStar           | 66     | 0      | 34           |
| RandomCommittee | 61     | 0      | 39           |
| Random Forest   | 60     | 0      | 40           |
| Random Tree     | 67     | 0      | 33           |
| Bounded K-means | 51     | 0      | 49           |

Table IX: Weka tool classifiers and Bounded k-means Clustering for less dominant emotion classification

others clusters. When we tried to classify these less dominant emotions (happiness, inspiration, sympathy, disgust, and fear), we found the following results presented in Table IX and Fig. 7.

#### E. User Identification

The raw measurements used for keystroke dynamics are dwell time and flight time. The time duration that a key is pressed is dwell time. The time duration between releasing a key and pressing the next key is flight time. While typing a series of characters, the time, the user needs to find the right key (flight time) and the time, he holds down a key (dwell time), is specific to that user, and can be calculated in such a way that it is independent of overall typing speed. Therefore, keystroke dynamics is unique for a specific user and can be

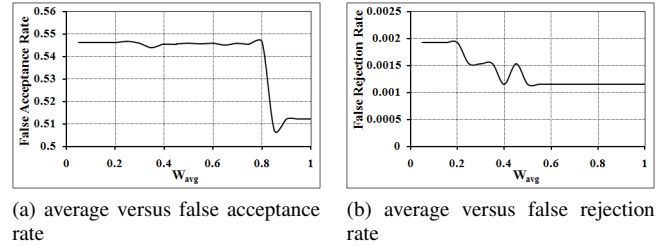


Figure 7: Percentage of average versus both false acceptance rate and false rejection rate in Bounded k-means for less dominant emotion classification

used for authentication. In the related work section, we have found that using keystroke dynamics for user authentication resulted in high accuracy. However, one of the constraints of user identification through keystroke dynamics, is identifying user in different emotional states.

Therefore, we attempted to identify user's identity based on the collected survey data after emotion induction. We analyzed the outcome of user authentication using keystroke dynamics, in both emotional state and neutral state. In this case, we considered the data without any normalization. At first, we attempted to cluster the data with our unsupervised clustering approach (Bounded K-means Clustering). However, we found it difficult to successfully identify users based on their usage attributes. The experimental results are as follows:

1) *Dominant user classification*: From matrix of users clusters, we found that some of the users heavily dominate others clusters. Therefore, we made an attempt to classify these dominant users. The results we found were much better than the general case. The results are illustrated in Fig. 8.

2) *Dominant and less dominant user classification with neutral emotion only*: In the above case, we attempted to classify the dominant users based on users' data for 10 emotional classes. However, these data can be different for different emotional states for the same user, making it harder to classify that user. Therefore, we took only one emotional state in consideration, which is neutral emotion and attempted to classify dominant and less dominant users by separate

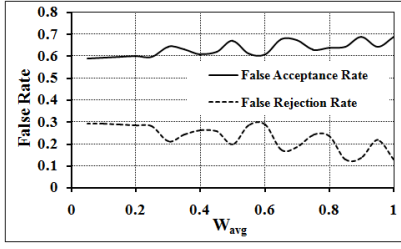


Figure 8: Percentage of average versus false accept rate and false reject rate in Bounded k-means for dominant user classification

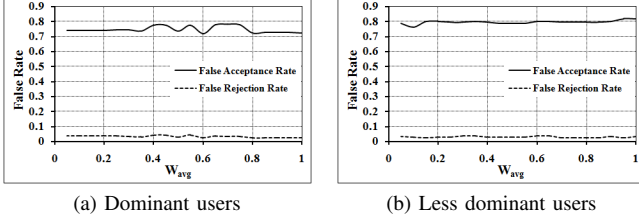


Figure 9: Percentage of average versus false acceptance rate and false rejection rate in Bounded k-means for dominant and less dominant user classification with neutral emotion data only

experimentation. The results are illustrated in Fig. 9.

#### IV. SUMMARY OF FINDINGS

We have found best accuracy of 69% in emotion detection using WEKA [5] tool classifier for 2-class emotion classification (positive and negative emotions) analysing on 3<sup>rd</sup> derivative of usage data. We also find that in our experiment, the clusters are highly overlapped with each other with dominant emotions data only and less dominant emotions data only, no emotion dominating its own cluster highly. The overall findings of our experiments for emotion detection is summarised in Table X. The summary demonstrates that our proposed classifier exhibit a moderate performance in classifying emotions compared to other existing classifiers. Besides, a noteworthy fact is that the performance of our proposed classifier does not deteriorate while segregating dominant and less dominant emotions. This does not hold true for other classifiers. Moreover, for the less dominant emotions, our proposed classifier performs the best.

The performance comparison over classification performance and operational overhead pertinent to WEKA [5] classifiers and our proposed classifier Bounded K-means Clustering is depicted in Table XI. Here, the classification time (total classifier running time - training time) is measured for per instance in milliseconds. Here, as per the measured values, our proposed mechanism demands extremely low resource compared to other alternatives and provides its output within a very short period of time.

It is worth mentioning that we have found several places in our study that we can attempt for further improvement. For

| Experiment                                                       | Best method            | Accuracy of the best method (%) | Accuracy of the Bounded K-means (%) |
|------------------------------------------------------------------|------------------------|---------------------------------|-------------------------------------|
| 10 classes of emotion                                            | Weka (Random Forest)   | 35                              | 23                                  |
| 2 classes of emotion (using original data)                       | Weka (KStar)           | 62                              | 49                                  |
| 2 classes of emotion (using 1 <sup>st</sup> derivative of data)  | Weka (RandomCommittee) | 63                              | 49                                  |
| 2 classes of emotions (using 2 <sup>nd</sup> derivative of data) | Weka (RandomCommittee) | 66                              | 50                                  |
| 2 classes of emotions (using 3 <sup>rd</sup> derivative of data) | Weka (RandomCommittee) | 69                              | 34                                  |
| Dominant 4 emotions only                                         | Weka (Random Forest)   | 53                              | 47                                  |
| Less dominant 5 emotions only                                    | Bounded K-means        | 49                              | 49                                  |

Table X: Summary of findings of emotion classification

| Method          | Classification time (ms) | CPU Usage (%) | Memory usage (MB) |
|-----------------|--------------------------|---------------|-------------------|
| Bounded K-means | 0.84                     | 40            | 26.5              |
| Bagging         | 5.88                     | 55            | 240               |
| IB1             | 1.17                     | 52            | 15                |
| KStar           | 191                      | 53            | 60                |
| RandomCommittee | 4.13                     | 72            | 345               |
| Random Forest   | 52.63                    | 59            | 480               |
| Random Tree     | 0.67                     | 37            | 110               |

Table XI: Comparison of classification performance and operational overhead of different classifiers used

example, in the end of each video clip, there is a related long paragraph for reading that may cause the induced emotion to fade away. In that case, this data will not be a correct representation of the induced emotional state. Besides, our survey system is a desktop application. That is why, we have failed to reach higher number of participants. Additionally the survey process is a lengthy one (around 40 minutes), which may be tiring for the participants. In case of user identification, our accuracy differs from the standard accuracy of identification using keystroke dynamics. It can be explained as follows: in different emotional states, individual keystroke pattern differs. Dominant user classification using Bounded K-means Clustering method resulted in 45% accuracy. However, dominant/less dominant user classification considering neutral emotions resulted in only 10% and 20% accuracy respectively

and user classification running different WEKA [5] classifiers on low arousal emotions resulted in 31% accuracy only.

## V. CONCLUSION

Devices, capable of detecting human emotion and interacting accordingly, is an important part of building intelligent computers. An emotionally-aware system will be much user-friendly and less frustrating for the users, as it will be able to make appropriate decisions about how to interact with the users or adapt their responses. There are two main problems with current system approaches for identifying emotions that limit their applicability: they demand additional infrastructures and they require specialized information, which are not always available. In this paper, to overcome both, we studied detecting emotion through analyzing the pattern of keyboard and mouse usage data. In our study, we attempted to identify user's different emotional states in diversified ways by using a newly proposed light-weight classifier namely Bounded K-means Clustering along with other existing classifiers such as KNN, KStar, RandomCommittee, Random Forest etc. We also attempted to identify users based on the usage data collected in different emotional states. Up to now, we have performed our study using a desktop version of our data collection system. In future, we plan to convert it into a web-based system. This will enlarge our data set and with the larger data set, we can investigate alternatives for further improvement of our analysis resulting in more accurate emotion detection.

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